

**DESCRIPTION OF DESIGN DATA FOR
COMMON STANDARD PARTS**

Part: **3.1 – MATERIAL DESIGN VALUES**

Chapter: **1350**

Title: **DESCRIPTION OF DESIGN DATA FOR
COMMON STANDARD PARTS**

Revision: **0.2**

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1 INTRODUCTION

This chapter presents the definitions of the abbreviations and symbols used in following chapters to refer, in stress and strength analyses, to design data of most common standard parts in use, including rivets, bolts and nuts.

1.1 DISCUSSION ON SOME GENERAL ASPECTS

The following discussion is temporally included in this chapter of the Stress Methods Manual. This inclusion herein is temporal; it shall be moved to a dedicated chapter in future revisions.

General

Materials Design Values to be used shall be consistent with applicable Stress Methods. In case a certain material property cannot be found in standard sources, or there are doubts about the applicability of some material data for a specific analysis, look for advice within your senior stress colleagues and/or the Stress Methods team.

Only materials fully qualified according to authorized specifications for aerospace applications will be allowed to be used in a part with any structural responsibility. For metallic materials and standard parts, this category includes the SAE/ASM specifications declared in the MMPDS, DIN/LN/WL specifications or Airbus specifications.

Statistical basis

General rules about the kind of statistical value of a material property that shall be used in analyses:

- Mean values of elastic moduli shall be used for internal loads/stress calculations

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- B-basis values of ultimate allowable stresses shall be used for strength analyses. S-basis values shall be typically acceptable if B-basis values not available. This last case (S-basis or requirement for minimum strength in specification) shall be the common case for standard parts as rivets, nuts, bearings, etc...

Data sources

Valid sources of DV data: MMPDS, HSB data-sheets, recognized international specifications (EN, NAS, ...), Airbus qualifications/specifications (ABS, DAN, CAN, ...) or Airbus Stress dedicated tests.

2 RIVETS AND BOLTS

Rivets and bolts, with their associated collar or nut when needed, conform the most common mean of fastening structural parts of airframes. Normally they will be selected from available standards instead of designed ad-hoc specifically for the structural detail.

Following tables describe the information needed for stress calculations, illustrated with values for a specific standard part as example. Dimensions in nominal/mean values preferably. If not specifically indicated in specification, minimum declared acceptable in the tolerance range.

2.1 REQUIRED DESIGN DATA FOR RIVETS AND BOLTS

Mandatory fields marked with (*)

IDENTIFICATION OF STANDARD PART

Name/designation (*)	EN6114T3	(1)
Version (*)	1.1 (12/2023)	
Description (*)	Hi-lite structural bolt with countersunk head	
Specifications (*)	EN6114	
Equivalences		
References		
Units	International System of Units (SI): subset of derived units (N,mm)	

(1) Including material and diameter code if data is stored individually for each rivet/bolt diameter size, and not for the whole "family"

GEOMETRY AND INSTALLATION DATA

Property	Symbol	Units	Value
Shank (nominal) diameter (*)	<i>d</i>	mm	4.80
Head type (*)	-	-	CSNK_100 (2)
Head outer diameter	<i>Dh</i>	mm	7.44
Head height	<i>Hh</i>	mm	1.42
Thread type	-	-	0.1900-32 UNJF-3A

(2) Label starting by character string PROT (for protruding head), or CSNK (for countersunk head) and, optionally, and underscore character followed by a marker/indicator of a main characteristic of the head type, eg. the opening angle in sexagesimal degrees of a countersunk head. Examples: PROT / CSNK / PROT_HEX / CSNK_100 / ...

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MATERIAL AND STRENGTHS

Property	Symbol	Units	Value
Designation of raw material (*)	-	-	Ti-6Al-4V
Young modulus of material	E	MPa	
Min. ultimate tensile strength of material	F_{tu}	MPa	
Min. ultimate shear strength of material	F_{su}	MPa	650
Single shear allowable load (*)	Q_{all}	N	11950
Tensile allowable load (*)	T_{all}	N	10700
Max. fatigue load (*)	Q_{fat}	N	4000 ⁽³⁾

⁽³⁾ "Max. fatigue load" is defined as the maximum single shear load in a monotonic fatigue spectrum of stress ratio R equal to 0.1 which will not cause failure (typically run-out of at least 1E6 cycles)

3 NUTS AND COLLARS

Nuts (for bolts) and collars (for Hi-Lok and Hi-Lite rivets) are parts required for completing the installation of most mechanical fastening systems. Information required for structural analysis presented in following tables in similar manner as done for bolts/rivets.

3.1 REQUIRED DESIGN DATA FOR NUTS AND COLLARS

IDENTIFICATION OF STANDARD PART

Name/designation (*)	ABS0258-6
Version (*)	1.2 (Dec-2023)
Description (*)	Collar Hi-Lok, type 303Se stainless steel
Specifications (*)	ABS0258
Equivalences	
References	
Units	International System of Units (SI): subset of derived units (N,mm)

GEOMETRY AND INSTALLATION DATA

Property	Symbol	Units	Value
Pin nominal diameter (*)	d	mm	4.80
Thread type	-	-	0.1900-32 UNJC-3B
Torque		Nm	2.82 - 3.95

MATERIAL AND STRENGTHS

Property	Symbol	Units	Value
Designation of raw material (*)	-	-	AISI 303Se
Tensile allowable load (*)	T_{all}	N	12232

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4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A-basis	-	Statistically-based material property set at the point where, with a 95% confidence, at least 99% of the population will exceed
B-basis	-	Statistically-based material property set at the point where, with a 95% confidence, at least 90% of the population will exceed
DV	-	Design Value, i.e. the value for a mechanical property of a structural material to be used in stress analyses
R	-	Stress ratio in a fatigue load cycle, i.e. ratio between minimum and maximum load in the cycle
S-basis	-	Specification minimum, or value based on specification minimum, of a material property

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 [SAE] MMPDS	>9	Metallic Materials Properties Development and Standardization

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.2 DD/01/2024	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

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APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods & Optimization</i>	G. Baños DD/01/2023	-
<i>Structural Analysis</i>	-	F. Glock DD/01/2023

Table 3. Approval signatures table for last revision of this chapter